Tengrui (Rio) Kong

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RESEARCH INTERESTS

Trustworthy and reliable machine learning for high-stakes domains, with emphasis on healthcare AI, scientific discovery, and LLM robustness. Specific interests: graph-based deep learning for neuroimaging, foundation model evaluation and adaptation in data-scarce settings, LLM faithfulness and misinformation robustness, and AI-assisted scientific reasoning.

EDUCATION

Incoming Ph.D. Student in Computer Science — Starting Aug 2026
Missouri University of Science and Technology - Rolla, MO — *Advisor: Dr. Seung-Jong Park*
Research focus: machine learning for biomedical and bioinformatics applications.

M.S. in Computer Science — Aug 2024 - May 2026
Missouri University of Science and Technology - Rolla, MO — *GPA: 3.81 / 4.0*
Selected coursework: Deep Learning, Natural Language Processing, Machine Learning, Advanced Topics in LLMs, Advanced Topics in Computer Vision, Analysis of Algorithms, Probability & Applications in Computing, Foundations of Data Management.

M.S. in Economics and Finance — 2015
University of International Business and Economics - Beijing, China

B.S. in Communication Engineering — 2012
East China Jiaotong University - Nanchang, China

PUBLICATIONS AND MANUSCRIPTS

[1] Tengrui Kong, Tao Wang, Minheng Chen, Ping Ma, Dajiang Zhu, Xiaowei Yu. Are Foundation Models Enough for Data-Scarce Functional Connectivity Diagnosis? Revisiting Task-Specific Learning with BrainLRR. *Under review, WACV 2027.*

[2] M. Alizade, Tengrui Kong, Suman K. Maity. SciTables: A Dataset and Evaluation Framework for Complex Table-to-Text Generation. *Accepted, VLDB 2026.* Contribution: dataset pipeline construction, LaTeX table extraction, LLM benchmarking, and fine-tuning.

SELECTED RESEARCH HIGHLIGHTS

- First-authored BrainLRR, a low-rank regularized graph Transformer for data-scarce functional connectivity diagnosis.
- Benchmarked task-specific models, vision backbones, LLMs, and neuroimaging foundation models across ASD, Alzheimer's disease, MCI, and Parkinson's disease datasets.
- Built reproducible PyTorch and SLURM-based HPC pipelines for multi-seed biomedical machine learning experiments.
- Investigated LLM robustness under politically charged misinformation and presupposition-based prompts.
- Developed LLM-guided optimization methods for astrophysics parameter estimation.

RESEARCH EXPERIENCE

Graduate Research Assistant — Dec 2025 - May 2026
Missouri S&T, CoBAI Lab - Rolla, MO — *Advisor: Dr. Xiaowei Yu*

- First-authored BrainLRR, a large-scale benchmark evaluating task-specific models and foundation models for data-scarce functional connectivity diagnosis across ASD, Alzheimer's disease, MCI, and Parkinson's disease datasets.
- Designed and trained a graph-based Transformer with stochastic node masking and low-rank regularization, improving ASD classification AUROC from 75.6% to 79.1% on resting-state fMRI functional connectivity.
- Benchmarked neuroimaging foundation models, including BrainLM and BrainSegFounder, as well as vision backbones such as ViT and ResNet, against task-specific baselines including Brain Network Transformer.
- Conducted ROI-level interpretability analysis to connect model predictions with clinically meaningful neuroimaging patterns.
- Built reproducible HPC training pipelines using PyTorch, SLURM, and the Hellbender cluster.

- GPU-accelerated a custom low-rank representation loss using torch.linalg.svd, achieving approximately 10x speedup over CPU-based SciPy SVD.

SELECTED RESEARCH PROJECTS

LLM-Guided Astrophysics Parameter Optimization Mar 2026 - May 2026

- Developed an LLM-guided optimizer for the 3ML + HAL HAWC astrophysics pipeline, replacing the Minuit minimizer with structured prompting and finite-difference gradient estimation.
- Reduced the log-likelihood gap from 1,760 to 38 after 200 iterations.
- Deployed the system locally using Ollama and Llama 3.1 to avoid external API costs and improve reproducibility.
- Implemented convergence diagnostics, stall detection, and parameter-bound clipping for robust optimization.

Political Misinformation Robustness in LLMs Sep 2025 - Dec 2025

- Constructed a benchmark using Wikipedia controversies and fact-checking archives to evaluate LLM robustness under politically charged prompts. Evaluated models including LLaMA-3.3-70B, Qwen-2.5-72B, GPT-OSS-120B, and Aya-Expanse-32B.
- Analyzed model behavior under presupposition-based prompts to study misinformation acceptance and reasoning robustness.

TECHNICAL SKILLS

Machine Learning and Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, graph neural networks, Transformer architectures.

LLMs and NLP: LLM fine-tuning, LoRA, QLoRA, supervised fine-tuning, prompt engineering, LLM-as-a-judge, RAG, BERT, SpaCy, NLTK, Ollama.

Neuroimaging and Biomedical AI: fMRI functional connectivity, nilearn, NetworkX, ABIDE, ADNI, PPMI, CC200 atlas, ROI-level analysis.

Infrastructure: SLURM, HPC, Hellbender cluster, AWS EC2, S3, RDS, Git, multi-GPU training.

Programming: Python, SQL, Java, JavaScript, C++.

INDUSTRY EXPERIENCE

Principal Research Specialist Nov 2020 - Jun 2024

Changsha Science & Technology Bureau - Changsha, China

- Applied quantitative analysis, Python, and SQL-based modeling to evaluate large-scale R&D portfolios and support technology policy decisions across 200+ companies.
- Designed statistical evaluation frameworks and forecasting dashboards for strategic planning and capital allocation.

Bond Investment Analyst, Fixed Income Department Jun 2015 - Nov 2020

First State Cinda Fund Management / Yingda Securities - Shenzhen, China

- Applied statistical modeling to credit risk assessment, fixed-income research, and portfolio analysis.
- Produced issuer reports and investment analyses supporting bond portfolio decisions.

CERTIFICATIONS

AWS Certified Cloud Practitioner - Issued Jul 2025; Expires Jul 2028